

Problem 5 Problem 5

Problem 1 (a) *True or False: If f and g are two functions defined on $(-1, 1)$, and if $\lim_{x \rightarrow 0} g(x) = 0$, then it must be true that $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = 0$.*

Explanation. False: Suppose

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and $g(x) = x$. Then $\lim_{x \rightarrow 0} g(x) = 0$ but

$$\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = \lim_{x \rightarrow 0} \left(\frac{1}{x} \cdot x \right) = \lim_{x \rightarrow 0} 1 = 1.$$

(b) *True or False: If f is continuous on $(-1, 1)$, and if $f(0) = 10$ and $\lim_{x \rightarrow 0} g(x) = 2$, then*

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 5.$$

Explanation. True: application of quotient rule. Because f is continuous, $\lim_{x \rightarrow 0} f(x) = f(0) = 10$

(c) *True or False: If f is continuous on $[1, 3]$, and if $f(1) = 0$ and $f(3) = 4$, then the equation $f(x) = \pi$ has a solution in $(1, 3)$.*

Explanation. True: f is continuous on $[1, 3]$, $f(1) = 0 < \pi < 4 = f(3)$, and the Intermediate Value Theorem implies there is some x in $(1, 3)$ with $f(x) = \pi$.

(d) *True or False: Let f be a positive function with vertical asymptote $x = 5$. Then*

$$\lim_{x \rightarrow 5} f(x) = \infty.$$

Explanation. False: Suppose f is defined by

$$f(x) = \begin{cases} \frac{1}{x-5} & \text{if } x > 5 \\ 2 & \text{if } x \leq 5. \end{cases}$$

Then f has a vertical asymptote at $x = 5$:

$$\lim_{x \rightarrow 5^+} \underbrace{\frac{1}{x-5}} = \infty,$$

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because the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator positive and goes to 0.

But, $\lim_{x \rightarrow 5^-} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 5} f(x)$ does not exist